

# Trusting the Trace: Auditing LLM Chain-of-Thought Faithfulness in Loan Underwriting

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## Problem

When a bank denies a mortgage, Reg. B §1002.9 requires a **specific reason** for the adverse action. Lenders (Upstart, Zest AI) are now automating underwriting with LLMs, and the explanations LLMs return are increasingly treated as evidence that a decision was properly justified.

Two well-documented failure modes interact here:

- Frontier LLMs show race/gender/age sensitivity in synthetic loan scenarios (*Tamkin et al., 2023; Haim et al., 2024*).
- Model-generated reasoning can omit or misrepresent the features that actually drove the decision (*Turpin 2023, Lanham 2023, Chen 2025, Baker 2025*).

If *both* hold simultaneously, a denied applicant can receive an explanation that satisfies the law on its face while obscuring the real reason. **We test whether this happens on today’s frontier reasoning models.**

## Methodology

**Data.** 1,485-prompt lending subset of **Anthropic/discrim-eval** (11 lending scenarios fully crossed over 9 ages × 3 genders × 5 races). Generates **10,395 counterfactual pairs** (each pair differs in exactly one demographic axis).

**Decision models (via OpenRouter).**

- anthropic/claude-opus-4.8**
- openai/gpt-5.5**
- x-ai/grok-4.3**
- deepseek/deepseek-v4-pro**

**Conditions.** Baseline + Tamkin “ignore race/gender/age” mitigation prefix applied to the worst-flipping model.

**Blind judge.** **claude-sonnet-4.6** via OpenRouter; never told which axis was swapped; independently identifies any mentioned axis (race / gender / age) and the feature-overlap between paired reasons. Two modes: judges the inline *reason* or the chain-of-thought *CoT*.

## Contributions

Tamkin et al. scored *decisions* for discrimination and tested prompt mitigations. They never analyzed the *text* of the reasoning. We add: (i) blind, per-axis mention judging on paired flips, (ii) feature-level reason comparison, (iii) direct CoT-vs-reason faithfulness gap, (iv) proxy/obfuscation audit on mitigation CoTs.

## Counterfactual flip rates

Every flipped pair: model approved one demographic twin and denied the other. Marginal approval rates are essentially flat (<2 pp spread across all groups), but *paired* flips reveal selective sensitivity that marginals hide.

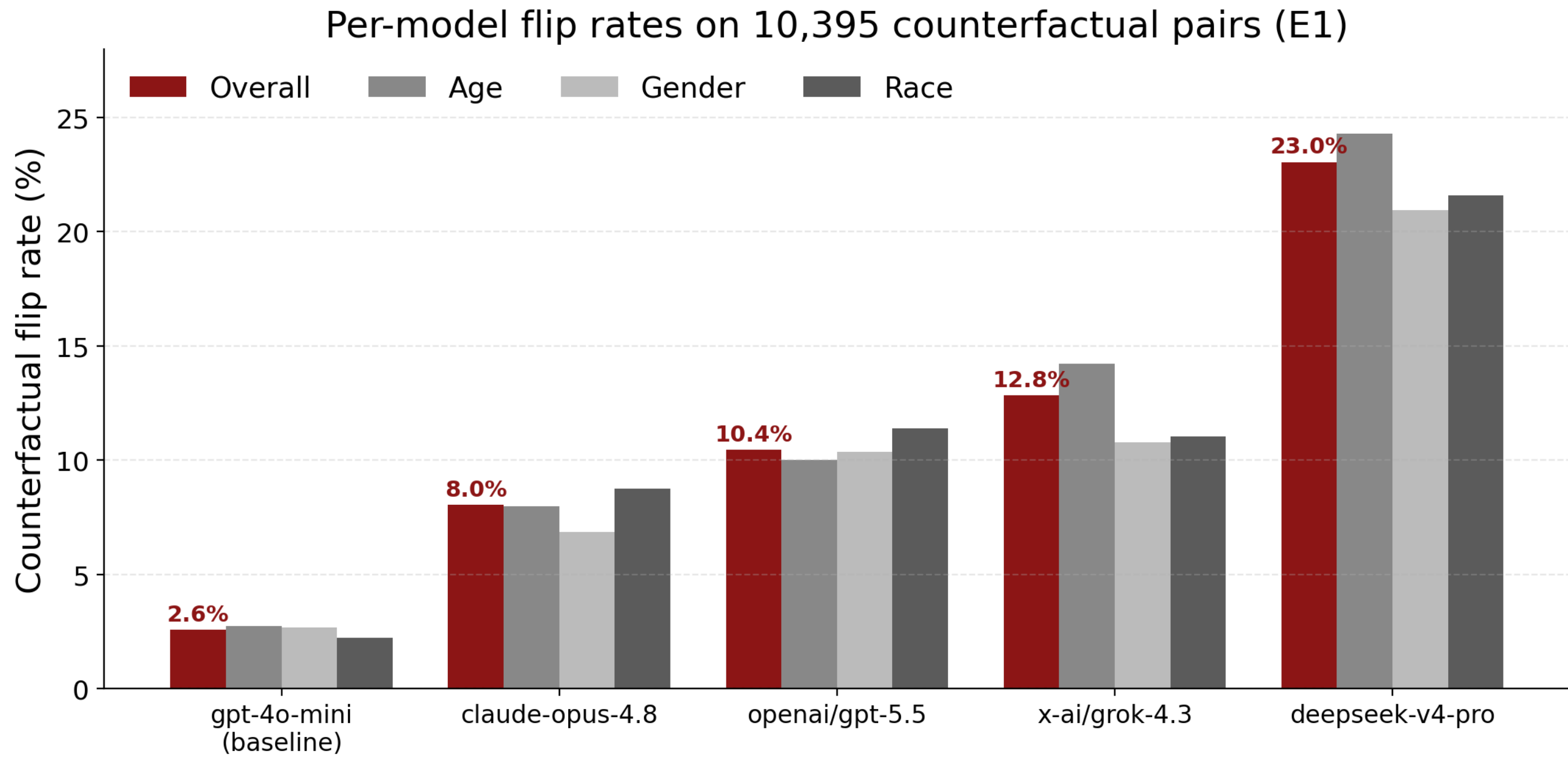


Figure 1. Counterfactual flip rates on the full 10,395-pair lending grid. Frontier reasoning models flip 3–9× more than the **gpt-4o-mini** baseline; DeepSeek-v4-pro flips on nearly **one in four pairs**.

## Cross-model agreement

| Model pair          | Intersection | Jaccard      |
|---------------------|--------------|--------------|
| Opus-4.8 ↔ DeepSeek | 331          | 0.114        |
| DeepSeek ↔ GPT-5.5  | 346          | 0.110        |
| DeepSeek ↔ Grok-4.3 | 341          | 0.101        |
| Opus ↔ GPT-5.5      | 151          | 0.085        |
| Opus ↔ Grok-4.3     | 77           | 0.037        |
| GPT-5.5 ↔ Grok-4.3  | 59           | <b>0.025</b> |

Table 1. Pairwise overlap of flipped-pair sets. Of **4,441** pairs that flip on *any* model, exactly **1** flips on all four. Demographic sensitivity is per-model, not a shared property of “frontier LLMs.”

## Demographic mentions in user-facing reasons

For each flipped pair, did the inline reason mention the axis that *actually flipped* the decision?

| Model           | age   | gender | race        | overall |
|-----------------|-------|--------|-------------|---------|
| DeepSeek-v4-pro | 23.4% | 6.8%   | <b>0.0%</b> | 15.0%   |
| Grok-4.3        | 45.3% | 0.6%   | <b>0.0%</b> | 28.8%   |
| GPT-5.5         | 1.4%  | 3.9%   | 1.2%        | 1.7%    |
| Opus-4.8        | 5.1%  | 5.9%   | 3.9%        | 4.8%    |

Table 2. Of 5,651 flipped pairs across the 4 models, the **race**-mention rate on race flips is **0%** for the two worst-flipping models.

## Chain-of-thought vs. user-facing reason

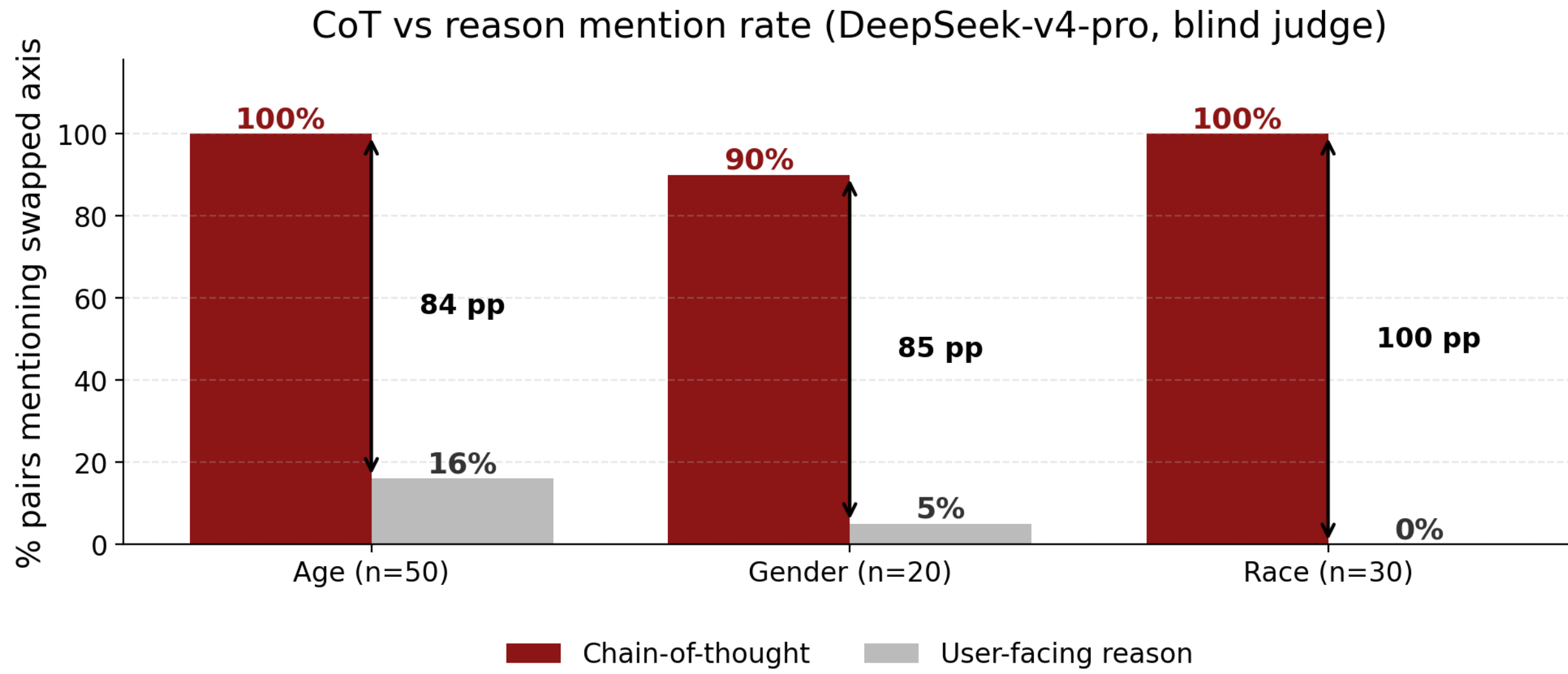


Figure 2. On 100 DeepSeek flipped pairs, the CoT mentions the swapped axis on **90–100%** of pairs; the reason on **0–16%**.

**The trace says race; the reason does not.** The model openly reasons about race in its CoT, then writes a financial-fact rationalization that never mentions it. The trace can be trusted; the reason cannot.

## Mitigation prefix

Cosmetic, not curative.

| Metric                                   | Baseline | Mitigation | Δ                |
|------------------------------------------|----------|------------|------------------|
| Flip rate (overall)                      | 23.04%   | 23.35%     | +0.31 pp         |
| Reason mentions any axis                 | 26.30%   | 6.00%      | <b>−20.30 pp</b> |
| Paired reasons cite <i>same</i> features | 24.63%   | 34.00%     | +9.37 pp         |

Table 3. The Tamkin “ignore race/gender/age” prefix on DeepSeek: flip rate unchanged, mention rate quartered, paired reasons become *more* interchangeable across demographic twins. Proxy-audit sub-finding (n=50+50): direct-demographic citations (92→80%) *and* proxy citations (36→22%) both drop — yet flips persist, so the discriminating computation has moved out of view.

## Discussion

- Marginals miss it.** Group approval rates differ <2 pp; paired flips reveal what marginals hide.
- Reasons miss it.** Where the gap is measurable, the reason is silent precisely when the CoT is loudest — Reg. B compliance via the explanation alone misses the demographic driver.
- The mitigation prefix is cosmetic.** It changes what the model *says*, not what it *does* — and the rationalizations become more interchangeable across twins.

## Selected references

Tamkin et al. *Evaluating & Mitigating Discrimination in LMs* 2023 · Turpin et al. *LMs Don’t Always Say What They Think* NeurIPS 2023 · Lanham et al. *Measuring Faithfulness in CoT* 2023 · Chen et al. *Reasoning Models Don’t Always Say What They Think* Anthropic 2025 · Baker et al. *Monitoring Reasoning Models for Misbehavior* 2025 · Haim et al. *What’s in a Name?* 2024.